

B. Tech. Degree VI Semester Special Supplementary Examination January 2017

ME 1603 OPERATIONS MANAGEMENT

(2012 Scheme)

Time: 3 Hours

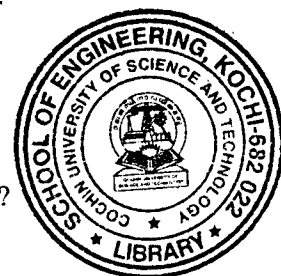
Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Define total float, free float and independent float.
 (b) Briefly explain the time series smoothening methods in demand forecasting.
 (c) Mention the main objectives of production planning and control.
 (d) Enunciate on bin cards and store ledger system.
 (e) Describe 'Gantt Chart'.
 (f) Explain aggregate planning.
 (g) What are the needs for plant layout?
 (h) What are the major selection factors for a suitable material handling system?



PART B

(4 × 15 = 60)

- II. (a) Defence organisation has come across with a new research and development programme. Which basic network techniques should be employed by them? Why? (5)
 (b) Table given below shows the normal time, crash time and incremental cost of crashing for various activities in a project. (10)

Activity	Nodes	Normal time (Days)	Crash time (Days)	Incremental cost of crashing (₹/day)
A	1-2	6	5	50
B	1-3	8	7	100
C	2-5	9	8	80
D	2-4	11	7	60
E	3-4	5	1	90
F	3-6	7	7	-
G	4-7	8	2	40
H	6-7	3	3	-
I	5-7	7	6	100
J	7-8	2	1	50

The total normal cost for performing 10 given activities = ₹2000, cost of supervision = ₹100 per day, penalty = ₹300 per day over 25 days, reward = ₹200 per day for less than or equal to 21 days.

Find (i) The total float for each activity.

(ii) Normal duration of the project and its related total cost.

(iii) The optimum duration and its minimum cost.

(iv) Minimum possible duration and its cost.

OR

- III. (a) For the given data find the forecast for each of the time period using simple moving average for $n = 3$ periods and weighted moving average for $n = 4$ periods. Also find the forecast for 9th and 10th period. (6)

Period	1	2	3	4	5	6	7	8
Demand	100	120	110	130	140	160	150	140

(P.T.O.)

- (b) The following are weekly sales data, in thousands of units, for micro-computer disks. (9)

113, 101, 98, 107, 120, 132, 110, 117, 112 and 125

Use $\alpha = 0.2$, $\alpha = 0.5$ and $\alpha = 0.8$ to produce an exponential smoothing model for these data and find the forecast for the 11th week. Using Mean Absolute Deviation (MAD) forecast error method and find out which value of α produces better forecast, assuming the initial forecast as 108.

- IV. (a) Explain the various stages of product life cycle. (7)

- (b) What are the objectives of inventory control? Also mention the various benefits for a production company in maintaining scientific inventory management. (8)

OR

- V. (a) Yantra India Ltd. is a supplier of speedometers to Speed Auto Ltd. manufacturers of 60 cc two-wheelers. It supplies 20,000 speedometers to Speed Auto Ltd annually. At Speed Auto, the ordering cost per order is ₹5 and the carrying cost is 2.5% of the average inventory value. The price of a single unit is ₹200. The company presently has a policy of placing 10 orders every year. Advise the management of Speed Auto Ltd as to whether it should continue with its present policy or switch to the EOQ model. (6)

- (b) Expected annual sales for a product is 8000 units and the ordering cost is ₹1800 per order. Holding cost is 10% of unit inventory value and the items can be purchased as given below. Determine the best order size. (9)

Quantity	Unit Price (₹/unit)
1-999	220
1000-1499	200
1500-1900	190
2000 and above	185

- VI. (a) Explain the graphical method used in aggregate planning. (8)

- (b) Write down the steps involved in sequencing of n jobs in m machines. (7)

OR

- VII. (a) Find the optimum sequence for the following set of job. Also find the minimum total elapsed time and idle time for each of the machines. Also prepare Gantt chart for both the machines. (7)

Jobs	A	B	C	D	E	F
M/c I	7	4	2	5	9	8
M/c II	3	8	6	6	4	1

- (b) Find an optimum sequence for the following set of jobs on six machines. Also find the total time elapsed and idle time for each machines. (8)

Jobs	Machines					
	A	B	C	D	E	F
1	19	8	8	3	11	24
2	18	6	9	6	9	18
3	12	5	8	5	7	15
4	20	5	3	4	8	11

- VIII. (a) Explain the various factors influencing plant location. (7)

- (b) Explain the different major material handling equipments with their adequate diagrammatic representations. (8)

OR

- IX. (a) Give detailed notes on process layout and product layout. (8)

- (b) Explain the classification of various industrial maintenance techniques in detail. (7)